

Contents

Chapter 7 — Sequences of Random Variables	1
7-1 General Concepts	1
Random Vectors, Joint Distribution, and Density	1
Independence and Repeated Trials	1
Mean, Covariance, and Correlation Matrices	2
7-2 Conditional Densities, Characteristic Functions, and Normality	2
Conditional Densities and Expectations	2
Characteristic Functions	3
Joint Normality	3
Complex Normal Vectors & Goodman's Theorem	3
Normal Quadratic Forms and Chi-Square Distribution	4

Chapter 7 — Sequences of Random Variables

7-1 General Concepts

Random Vectors, Joint Distribution, and Density

A random vector $X = [x_1, \dots, x_n]$ is a vector whose components x_i are random variables. * **Joint Distribution (CDF):** $F(X) = P\{x_1 \leq x_1, \dots, x_n \leq x_n\}$. * **Joint Density (PDF):** $f(X) = \frac{\partial^n F(x_1, \dots, x_n)}{\partial x_1 \dots \partial x_n}$. The probability that X is in a region D is $P\{X \in D\} = \int_D f(X) dX$. * **Marginals:** The joint density of a subset of variables is obtained by integrating out the remaining variables from the full joint density. * **Transformations:** For a system of functions $y_i = g_i(X)$, the joint density of $Y = [y_1, \dots, y_n]$ is:

$$f_y(y_1, \dots, y_n) = \frac{f_x(x_1, \dots, x_n)}{|J(x_1, \dots, x_n)|}$$

where J is the Jacobian determinant of the transformation.

Example 7-1 (Sum of Variables): Given n independent random variables x_i with densities $f_i(x_i)$, form $y_k = x_1 + \dots + x_k$ for $k = 1, \dots, n$. To find the joint density of y_k , we solve the system $x_1 = y_1, x_1 + x_2 = y_2, \dots, x_1 + \dots + x_n = y_n$. The unique solution is $x_k = y_k - y_{k-1}$. The Jacobian equals 1, yielding:

$$f_y(y_1, \dots, y_n) = f_1(y_1)f_2(y_2 - y_1) \dots f_n(y_n - y_{n-1})$$

Independence and Repeated Trials

- **Independence:** Random variables are mutually independent if their joint distribution/density factors into the product of their marginals: $f(x_1, \dots, x_n) = f(x_1) \dots f(x_n)$.
- **Repeated Trials:** Repeating an experiment n times generates n independent, identically distributed (i.i.d.) random variables.

Example 7-2 (Order Statistics): Let x_i be n i.i.d. variables. Ordering their values x_i yields $y_1 \leq y_2 \leq \dots \leq y_n$. The density of the k -th order statistic y_k is:

$$f_k(y) = \frac{n!}{(k-1)!(n-k)!} F_x^{k-1}(y) [1 - F_x(y)]^{n-k} f_x(y)$$

Special Case: If x_i are exponential with parameter λ ($f_x(x) = \lambda e^{-\lambda x} U(x)$), their minimum y_1 has the density $f_1(y) = n\lambda e^{-n\lambda y} U(y)$, which is also exponential with parameter $n\lambda$.

Example 7-3 (Failure Rate): A system consists of m components. Let x_i be the time to failure of the i -th component, with common distribution $F(t)$. Let $n(t)$ be the number of components good at time t . The

expected number of good components is $\eta(t) = E\{n(t)\} = m[1 - F(t)]$. The relative expected failure rate is defined as:

$$\beta(t) = -\frac{\eta'(t)}{\eta(t)} = \frac{f(t)}{1 - F(t)}$$

Example 7-4 (Measurement Errors): An object of length η is measured n times, yielding $x_i = \eta + v_i$. The noise terms v_i are independent with zero mean and variance σ_i^2 . We seek an unbiased linear estimate $\hat{\eta} = \sum_{i=1}^n \alpha_i x_i$ that minimizes variance $V = \sum_{i=1}^n \alpha_i^2 \sigma_i^2$, subject to $\sum \alpha_i = 1$. Using Lagrange multipliers, the optimal estimate is:

$$\hat{\eta} = \frac{\sum_{i=1}^n x_i / \sigma_i^2}{\sum_{i=1}^n 1 / \sigma_i^2}$$

Mean, Covariance, and Correlation Matrices

- **Linearity:** The expected value operator is linear: $E\{\sum a_i g_i(X)\} = \sum a_i E\{g_i(X)\}$.
- **Covariance:** $C_{ij} = E\{(x_i - \eta_i)(x_j^* - \eta_j^*)\} = E\{x_i x_j^*\} - E\{x_i\}E\{x_j^*\}$.
- **Uncorrelated Variables:** If $C_{ij} = 0$ for $i \neq j$, variables are uncorrelated. For uncorrelated variables, the variance of the sum is the sum of the variances: $\sigma_x^2 = \sigma_1^2 + \dots + \sigma_n^2$.

Example 7-5 (Sample Mean and Variance): For uncorrelated RVs with identical mean η and variance σ^2 : * Sample mean: $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i \implies E\{\bar{x}\} = \eta, \sigma_{\bar{x}}^2 = \frac{\sigma^2}{n}$. * Sample variance: $\bar{v} = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$. It is an unbiased estimator: $E\{\bar{v}\} = \sigma^2$.

- **Correlation Matrix (R_n):** The matrix with elements $R_{ij} = E\{x_i x_j^*\}$.
 - **Theorem 7-1:** R_n is a nonnegative definite matrix ($AR_n A^\dagger \geq 0$).
 - Its determinant $\Delta_n \geq 0$, with $\Delta_n = 0$ if and only if the variables are linearly dependent.

7-2 Conditional Densities, Characteristic Functions, and Normality

Conditional Densities and Expectations

- **Conditional Density:** $f(x_n, \dots, x_{k+1} | x_k, \dots, x_1) = \frac{f(x_1, \dots, x_n)}{f(x_1, \dots, x_k)}$.
- **Chain Rule:** $f(x_1, \dots, x_n) = f(x_n | x_{n-1}, \dots, x_1) \dots f(x_2 | x_1) f(x_1)$.
- **Chapman-Kolmogoroff Equation:**

$$f(x_1 | x_3) = \int_{-\infty}^{\infty} f(x_1 | x_2, x_3) f(x_2 | x_3) dx_2$$

- **Law of Total Expectation:** $E\{E\{x_1 | x_2, \dots, x_n\}\} = E\{x_1\}$.

Example 7-6 (Transforming to Uniforms): Given n arbitrary random variables x_i , we construct $y_1 = F(x_1)$, $y_2 = F(x_2 | x_1)$, $\dots$, $y_n = F(x_n | x_{n-1}, \dots, x_1)$. The Jacobian determinant is triangular with diagonals $\frac{\partial y_k}{\partial x_k} = f(x_k | x_{k-1}, \dots, x_1)$. Applying the transformation rule and chain rule yields:

$$f(y_1, \dots, y_n) = \frac{f(x_1, \dots, x_n)}{f(x_1) f(x_2 | x_1) \dots f(x_n | x_{n-1}, \dots, x_1)} = 1$$

in the cube $0 \leq y_i \leq 1$, meaning the y_i are independent uniform variables.

Random Sums Example (Eq 7-46 to 7-49): Let $s = \sum_{k=1}^n x_k$, where n is a discrete random variable independent of the i.i.d sequence x_k . * $E\{s\} = E\{E\{s|n\}\} = E\{\eta n\} = \eta E\{n\}$. * $E\{s^2\} = E\{E\{s^2|n\}\}$.

Since $E\{s^2|n\} = \eta^2 n^2 + \sigma^2 n$, we get $E\{s^2\} = \eta^2 E\{n^2\} + \sigma^2 E\{n\}$. * *Special Case:* If n is a Poisson variable with parameter λt (emitted particles) and x_k is particle energy (Maxwell distribution with mean $3kT/2$), total energy s has mean $E\{s\} = \frac{3kT\lambda t}{2}$.

Characteristic Functions

- **Definition:** $\Phi(\Omega) = E\{\exp(j\Omega X^t)\} = E\{\exp(j(\omega_1 x_1 + \dots + \omega_n x_n))\}$.
- **Sum of Independent RVs:** For $z = x_1 + \dots + x_n$, the PDF is the convolution of the individual PDFs: $f_z(z) = f_1(z) * \dots * f_n(z)$. Thus, the characteristic function factors:

$$\Phi_z(\omega) = \Phi_1(\omega) \dots \Phi_n(\omega)$$

Example 7-8 (Bernoulli & Poisson): (a) Let $x_i = 1$ with probability p , and 0 with probability q . The characteristic function is $\Phi_i(\omega) = pe^{j\omega} + q$. The sum $z = \sum x_i$ counts successes in n trials. $\Phi_z(\omega) = (pe^{j\omega} + q)^n$, verifying the Binomial distribution. (b) If $p_i \ll 1$ and $p_1 + \dots + p_n \rightarrow a$, then $pe^{j\omega} + q \simeq e^{p_i(e^{j\omega} - 1)}$. The product $\Phi_z(\omega) \simeq e^{a(e^{j\omega} - 1)}$, which is the characteristic function of the Poisson distribution.

Joint Normality

- **Definition:** Random variables $x_1, \dots, x_n$ are jointly normal iff any linear combination $w = a_1 x_1 + \dots + a_n x_n$ is a normal random variable.
- **Characteristic Function:** For a zero-mean normal vector with covariance matrix C :

$$\Phi(\Omega) = \exp\left\{-\frac{1}{2}\Omega C \Omega^t\right\}$$

- **Joint Density:**

$$f(X) = \frac{1}{\sqrt{(2\pi)^n \Delta}} \exp\left\{-\frac{1}{2}X C^{-1} X^t\right\}$$

where $\Delta = \det(C)$.

- **Important Property:** If jointly normal random variables are mutually uncorrelated (i.e., their covariance matrix is diagonal), they are also mutually independent.

Example 7-9 (4th Moment of Joint Normals): For jointly normal RVs with zero mean, expanding the characteristic function exponentials and equating coefficients yields:

$$E\{x_1 x_2 x_3 x_4\} = C_{12}C_{34} + C_{13}C_{24} + C_{14}C_{23}$$

Complex Normal Vectors & Goodman's Theorem

- For a complex random vector $Z = X + jY$, the density generally requires specifying $2n^2 + n$ parameters for the real and imaginary parts.
- **Goodman's Theorem:** If $C_{XX} = C_{YY}$ and $C_{XY} = -C_{YX}$, the complex vector is statistically determined solely by its complex covariance matrix $C_{ZZ} = 2(C_{XX} - jC_{XY})$. The joint density is simply:

$$f_Z(Z) = \frac{1}{\pi^n |C_{ZZ}|} \exp\{-Z C_{ZZ}^{-1} Z^\dagger\}$$

Normal Quadratic Forms and Chi-Square Distribution

[Image of chi-square distribution]

- **Theorem:** Given n independent $N(0, 1)$ random variables z_i , the sum of their squares $x = z_1^2 + \cdots + z_n^2$ follows a $\chi^2(n)$ distribution.
- If $x \sim \chi^2(m)$ and $y \sim \chi^2(n)$ are independent, then their sum $z = x + y \sim \chi^2(m + n)$.

Example 7-10 (Sample Variance of Normal Variables): Given n i.i.d. $N(\eta, \sigma)$ variables x_i , we form the sample variance $s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$. Summing the identity $(x_i - \eta)^2 = (x_i - \bar{x})^2 + (\bar{x} - \eta)^2 + 2(x_i - \bar{x})(\bar{x} - \eta)$ yields:

$$\sum_{i=1}^n \left(\frac{x_i - \eta}{\sigma} \right)^2 = \sum_{i=1}^n \left(\frac{x_i - \bar{x}}{\sigma} \right)^2 + n \left(\frac{\bar{x} - \eta}{\sigma} \right)^2$$

The left side is $\chi^2(n)$. The last term is $\chi^2(1)$ because $\bar{x}$ is $N(\eta, \sigma/\sqrt{n})$. Because $\bar{x}$ and s^2 are independent, it follows that $\frac{(n-1)s^2}{\sigma^2}$ is a $\chi^2(n-1)$ random variable.